Advocacy and Data Collection: Graduate Students' Association Cost-of-Living Survey

Aashish Kumar
1870378

Supervisor
Dr. Michael Li

University of Alberta, Edmonton, AB

For the partial fulfillment of the requirements for the M.Sc. in Mathematical and Statistical Sciences

Acknowledgement

I am deeply grateful to the graduate students who took the time to complete the Cost of Living Survey and share their experiences. I appreciate having your candour and the trust you placed in the Graduate Students' Association to use your responses anonymously and responsibly for advocacy.

I also want to thank the Graduate Students' Association team and executives for supporting the development, distribution, and administration of the survey, and for providing guidance on the priorities which shaped this report. I am grateful to the departmental and faculty graduate associations, student representatives, and campus partners who helped circulate the survey and encouraged participation.

Finally, I appreciate everyone who reviewed drafts, offered feedback on the analysis and presentation, and helped strengthen the clarity plus usefulness of this capstone.

Abstract

This capstone report presents the design, administration, and analysis of a Graduate Students' Association cost of living survey, which addresses an evidence gap regarding graduate student expenses, basic needs insecurity, and funding adequacy. The survey also aims to generate advocacy ready indicators for discussions with the University of Alberta and external stakeholders. Analysis of 1,026 anonymous survey submissions involved standardising mixed response formats into consistent measures of monthly housing, food, and transportation costs. An Essential Cost Index (housing plus food plus transportation) and a housing share indicator were constructed, and hardship was summarised using both a direct food insecurity item and an agree-disagree severity scale. Essential monthly costs were high for many respondents, with a median Essential Cost Index of $1,960 per month (interquartile range $1,550 to $2,300) among those with complete essential cost data. Nearly half of respondents who answered the direct item reported experiencing some level of food insecurity (46.6 per cent). An exploratory logistic regression linking cost burden measures to food insecurity indicated that housing share was strongly associated with higher odds of food insecurity (odds ratio 5.76, 95 per cent confidence interval 1.52 to 21.85). International student status was associated with substantially higher odds (odds ratio 2.59, 95 per cent confidence interval 1.88 to 3.58), while higher annual funding was associated with lower odds of food insecurity (odds ratio 0.98 per additional $1,000 of funding). These outcomes uphold the report's core message that housing cost dominance and funding inadequacy are central drivers of insecurity in basic needs for many graduate students. The results support recommendations to align funding with local cost pressures, strengthen housing and food supports, and track repeatable indicators over time, while accepting the limitations of voluntary response sampling and self-reported expenses.

Table of Contents

1. Introduction

Graduate students at the University of Alberta face increasingly diverse financial challenges, including rising housing costs, limited stipends, unpredictable living expenses, and unequal access to support systems. These pressures directly affect their health and academic performance. Although these concerns are significant for university policy and graduate student advocacy, there is a lack of routinely collected, student-specific cost-of-living data. During my term as President of the Graduate Students' Association (GSA), I initiated and led a focused effort to address this evidence gap by designing, administering, and analysing a cost-of-living survey for graduate students. This report documents the internship project, including its motivation, research questions, data collection process, analytic methods, principal findings, and recommendations for ongoing measurement and policy action.

This project generated an initial dataset and summary indicators that the GSA can use in negotiations with the university and external stakeholders (i.e. the Government of Alberta) to advocate for support in the most burdensome expense categories. From both academic and professional perspectives, the project fulfils the MDP internship requirement by translating policy-relevant questions into a structured data-collection and analysis plan, coordinating stakeholders throughout execution, and producing actionable findings.

The project aimed to achieve the following objectives:

1. Develop a concise, student-centred survey instrument to measure monthly and episodic expenses across key categories, including housing, food, utilities, tuition and fees, transit, healthcare, and others.
2. Collect responses from a representative cross-section of graduate students.
3. Produce descriptive cost-of-living indicators and conduct basic comparative assessments by funding type, domestic or international status, and program grouping.
4. Translate the findings into a set of prioritised advocacy recommendations for the GSA.

To achieve these objectives, I led survey design, managed ethical and consent procedures and communications, conducted data cleaning, and presented results to GSA leadership and relevant campus offices. The remainder of the report is organised as follows: Section 2 situates the project within the context of existing measures and student needs. Section 3 outlines the project objectives and practical questions. Section 4 describes my role and the sequence of activities. Sections 5 and 6 detail the data, collection methods, and analysis. Section 7 presents the results. Sections 8 through 10 provide interpretation, recommendations, and conclusions. The appendices include the survey instrument, data dictionary, cleaning log, and supporting documents.

2. Background and Literature

Graduate students in Canada are experiencing increasing cost-of-living pressures on basic needs, mainly due to post-pandemic inflation, stagnant funding, and rising student fees. By April 2022, Statistics Canada reported that the Consumer Price Index had risen by approximately 6.8% nationwide, with food prices increasing by 9.7% (Statistics Canada, 2022). These increases have outpaced the growth in living costs for many graduate students. A recent national survey indicated that “the majority of graduate students are facing serious financial concerns,” primarily as a result of stagnant stipends and growing expenses (Laframboise et al., 2023). The University of Alberta introduced a minimum total funding package of $25,000 per year (pre-tuition) in fall 2025 (University of Alberta, 2024), placing most students at or below the national poverty threshold. In Edmonton, this amount is significantly below the local living wage, estimated at approximately $42,400 per year (Alberta Living Wage Network, 2024).

2.1. Housing Costs

Housing in Canada remains prohibitively expensive, particularly for young adults. In late 2024, Statistics Canada reported that nearly 45% of Canadians were “very concerned” about their ability to afford housing or rent (Statistics Canada, 2024). This concern is especially pronounced among individuals aged 20 to 35, with 59% reporting significant anxiety about housing affordability and over half indicating that rising prices had altered their moving plans (Statistics Canada, 2024). These problems result in high rent-to-income ratios. According to the Canadian Alliance of Student Associations (CASA), 72% of students reported spending 30% or more of their income on housing, surpassing the standard threshold for unaffordability (Canadian Alliance of Student Associations, 2023). The situation is particularly acute in university towns, where Canada Mortgage and Housing Corporation (CMHC) data indicate historically low vacancy rates (approximately 2.2% nationwide in 2024) and elevated rents in major cities (Ronson, 2025). For example, Vancouver's vacancy rate is only 1.6%, with average two-bedroom rents at $2,173 per month (Ronson, 2025). In Edmonton, an adult monthly transit pass costs about $102, resulting in annual commuting expenses exceeding $1,200 (City of Edmonton, 2026). Consequently, most student households spend well above 30% of their income on housing (Canadian Alliance of Student Associations, 2023), indicating a broader trend: over one in five Canadian households lived in “unaffordable” housing in 2022 (Statistics Canada, 2024). In Alberta, although rental supply has increased modestly, improving vacancy rates, housing remains a primary source of stress for students in Edmonton and Calgary, both of which continue to experience low vacancy rates (Ronson, 2025).

2.2. Food Insecurity

Food insecurity is prevalent among post-secondary students, though estimates vary by measurement method. A recent population-based study found that 15.0% of full-time and 16.2% of part-time young post-secondary students experienced food insecurity in the past year (Wang et al., 2023). These rates are slightly lower than the 19.2% observed among similarly aged non-students (Wang et al., 2023). However, student associations and campus-based surveys report substantially higher rates. For instance, a 2023 national CASA survey found that nearly 39% of students reported experiencing some level of food insecurity in the previous year (Canadian Alliance of Student Associations, 2023). Other campus studies have indicated that over half of surveyed students are food-insecure, with one Fall 2021 multi-campus survey reporting 56.8% as moderately or severely insecure (Wang et al., 2023). Disparities are evident based on circumstances: students with children, those living in rented housing, or those relying on social assistance face significantly higher odds of food insecurity (Wang et al., 2023). In Alberta, these patterns reflect the national situation. Prairie households are contending with rising food prices; inflation reached approximately 8.8% in mid-2022 (Uppal, 2023), and student food banks continue to report high demand. These findings reveal the co-occurrence of food and financial insecurity, as students frequently reduce grocery spending to favour rent or tuition payments.

2.3. Tuition and Academic Costs

Tuition and academic fees represent major financial burdens for graduate students. Nationally, Canadian graduate tuition has been increasing gradually; Statistics Canada projects that average full-time graduate tuition will reach $7,978 in 2025/26, up 0.9% from the previous year. Significant interprovincial variation exists: Newfoundland and Labrador ($4,081) and Quebec ($4,307) have the lowest graduate fees for domestic students, whereas Nova Scotia and British Columbia are expected to exceed $10,000. Alberta's graduate fees are close to the national average, attributed to moderate tuition subsidies. In contrast, international graduate students are projected to pay much higher fees, averaging $24,028 in 2025/26 (Statistics Canada, 2025). Beyond tuition, students are required to pay mandatory fees, typically several hundred dollars annually, and to purchase textbooks and academic supplies. For example, national student financial surveys report that undergraduate students spend approximately $1,000–$1,500 per academic year on textbooks and supplies (Statistics Canada, 2024), while institutional estimates indicate that mandatory non-instructional student fees can range from approximately $800 to over $1,500 annually, depending on the university and program (University of Alberta, 2024). These non-tuition academic expenses, which are frequently not covered by scholarships or bursaries, can increase a student's annual budget by several thousand dollars. Tuition relief and grant availability also vary; for instance, Alberta Student Aid permits annual loan limits up to $34,000 for most graduate programs, indicating the substantial costs involved (Alberta Student Aid, n.d.).

2.4. Healthcare Expenses

Although Canada's public healthcare system provides essential physician and hospital services to residents, students still face out-of-pocket health expenses. In 2023, the average Canadian household spent approximately $3,087 on healthcare, including prescription drugs, dental and vision care, and other medical supplies (Statistics Canada, 2025). While many students are exempt from provincial health premiums, they frequently incur costs for dental care, medications, eye care, and private insurance. For instance, doctoral students at numerous universities are required to enrol in mandatory student health plans, which cost several hundred dollars annually. These health-related expenses are in addition to other living costs. Notably, a national survey found that approximately 17.5% of Canadians incurred debt to pay for healthcare expenses, primarily for dental and prescription costs (National Institute on Ageing, 2024).

Although no Canada-wide statistic focuses exclusively on students, unexpected medical or dental expenses are likely to add further strain to already limited student budgets.

2.5. Transportation Costs

Transportation accounts for a significant portion of students' expenses. In 2023, transportation accounted for 15.8% of average household spending, with households spending an average of $12,090 annually on transportation, including vehicle costs and public transit (Statistics Canada, 2025). Students residing off campus frequently depend on public transit or personal vehicles. For example, an adult monthly transit pass in Edmonton costs approximately $102, amounting to over $1,200 per year (City of Edmonton, 2026). Car ownership, encompassing insurance, fuel, and maintenance, can result in even higher costs. These transportation expenses compete with other essential living costs, and many students report allocating a substantial portion of their income to commuting. In Alberta’s cities, where car travel is prevalent, students may need to budget several thousand dollars annually for transportation.

To summarise, graduate students in Canada experience considerable cost-of-living challenges. Housing and rent frequently consume more than 30% of student income (Canadian Alliance of Student Associations, 2023). Food insecurity impacts a significant minority of students, with prevalence estimates ranging from 15% to 40% (Elgar et al., 2023). Tuition and fees, which average between $5,000 and $10,000 or more annually, depending on the province (Statistics Canada, 2025), add to academic costs, along with expenses for textbooks and supplies. Additionally, mandatory healthcare (including vision, dental, and prescription costs) and transportation expenses further increase monetary burdens (Statistics Canada, 2025). These factors are cumulative; surveys indicate that students often must choose between paying rent and purchasing groceries or covering other bills. The literature, therefore, identifies a complex affordability crisis among Canadian graduate students, highlighting the need for government interventions, such as increased funding, subsidised housing, food assistance, and accessible transit passes, to alleviate these financial pressures.

3. Objectives and Practical Questions

3.1. Purpose of the project

The capstone project aimed to create credible, student centred cost-of-living indicators reflecting real graduate student expenses and their interactions with funding, workload, and basic needs. These will support GSA advocacy and university planning, offering more relevant data than general inflation or household indices. The aim is to produce methodologically transparent, practical evidence directly usable in decision making, supporting discussions with both university leadership and external stakeholders such as the Board of Governors.

3.2. Core Objectives

This project pursued three interconnected objectives.

3.2.1 The first objective was to design a concise, policy-relevant survey instrument tailored to the economic circumstances of graduate students. This was necessary because existing surveys and university or government data do not capture the cost realities or provide the granular, student centred insights needed for effective support and advocacy. By focusing on the main components of their cost of living, housing, food, utilities, transportation, tuition and fees, healthcare, and other academic expenses, the survey enables a targeted understanding of challenges unique to this group.

3.2.2 Collect and curate data that shows graduate student expenses and financial situations across program type, faculty, enrollment status, residency, and presence of dependents.

3.2.3 Construct clear, interpretable indicators of cost burden and monetary strain, including expense summaries, and translate findings into evidence for GSA advocacy, negotiations, planning, and program design.

3.3. Practical Questions Guiding Survey Design and Analysis

The objectives were operationalised through practical, policy-oriented questions. These guided both the instrument design and the analytical approach.

3.3.1. What does the typical graduate student spend each month across major expense categories?

This includes housing (rent and housing-related costs), food, transportation, utilities, tuition and fees (where reported), healthcare costs, and other necessary expenses. The focus is on typical values (medians) and the spread of outcomes (distributions), since policy must address both central tendencies and high-burden tails.

3.3.2. Which expense categories create the greatest pressure on students' budgets?

The analysis highlights leading expense categories that strain budgets, informing cost-offset policies, funding, and support.

3.3.3. What proportion of respondents experience high monetary strain or basic-needs insecurity?

Financial hardship is captured through indicators such as whether funding covers day-to-day expenses, the need for additional money, coping strategies (e.g., loans, credit, family support), and basic needs outcomes (e.g., food insecurity and use of supports, such as food bank utilisation).

3.3.4. Do cost-of-living experiences differ across key groups of students?

The analysis compares residency, program, enrollment, year, faculty, and dependents to identify which groups face the greatest burdens, supporting equity focused interpretation.

3.3.5. Which areas of spending and financial circumstance are most closely associated with hardship outcomes?

The analysis links cost categories and burden indicators to hardship outcomes, such as food insecurity, and to coping strategies, thereby helping to prioritise support for the most affected areas.

3.3.6. What interventions do students prioritise when asked directly?

Student rankings and support preferences reveal which initiatives are seen as most helpful, aligning needs with student priorities.

3.4. Understanding how these questions connect to intended outputs

The project delivers actionable outputs: cost summaries by category, indicators such as essential-cost and housing share measures, subgroup comparisons for targeted recommendations, and ranked priority lists linking student needs and advocacy, including support for graduate student parents and dependents. Together, these objectives and questions ensure the work is not only descriptive but also decision oriented: it produces evidence that can support policy talks, budget negotiations, and the prioritisation of student support programs.

4. Data Collection Methods

4.1. Dataset and Response Window

The analysis adopts the UofA GSA Cost of Living Survey data file, which contains 1,026 responses and 53 survey variables. Each row represents a single survey submission, and each column relates to a specific survey item. The response window is determined by the timestamp recorded for each submission in the exported dataset.

The survey was widely distributed, and participation was voluntary. As of 1 December 2025, the total number of graduate students was 8,344 (University of Alberta, 2025). With 1,026 survey responses, the response rate is approximately 12.3%. Based on the response rate, the dataset is still best treated as a voluntary response sample, suitable for describing respondents' experiences and identifying cost-pressure patterns, rather than for generating population-representative prevalence estimates.

4.2. Survey Platform and Administration

The survey was administered using Google Forms. The instrument introduction stated that the purpose was to gather information about cost of living problems faced by graduate students to support advocacy within the university and at provincial and federal levels. The introduction also indicated an expected completion time of approximately five to six minutes, which informed the decision to keep the instrument compact while still addressing major cost and hardship domains.

4.3. Survey Instrument Development

The survey instrument was designed to generate graduate student centred indicators showing both monthly expenses and common forms of monetary vulnerability during graduate study. Question development followed an iterative process based on a review of comparable student cost of living and basic needs surveys from other universities, as well as informal consultation with colleagues. Additional desk research refined the wording and response options to guarantee clarity for respondents and straightforward applicability for policy-oriented reporting.

The final instrument pairs expense measures with academic and demographic context variables to enable subgroup comparisons and includes items that record priorities for institutional supports.

4.4. Target Population, Recruitment, and Participation Incentive

The target population consisted of graduate students at the University of Alberta across all faculties and program types. The survey was conducted via email invitations sent to graduate students' email addresses, aiming to reach a broad cross-section of the graduate student body.

Participation was voluntary. To encourage engagement, the survey included a prize draw for two tickets to the Edmonton Oilers versus Calgary Flames game. This incentive was optional and did not compromise the anonymity of survey responses, as entry into the draw was separated from the main response process.

4.5. Survey Structure and Measures Collected

The survey was organised into themed modules which correspond with the project's analytical objectives.

4.5.1. Demographic and scholarly context

Items captured include degree program, enrolment status, primary source of financial support, faculty, year of study, and residency status in Canada. These variables facilitate comparisons across key groups in the analysis.

4.5.2. Food expenses and food insecurity

The food section includes a categorical monthly food expense item and a direct food insecurity item. It also features an agree-or-disagree question covering multiple experiences of food insecurity and a top-three ranking question on the food security initiatives most beneficial to students. The section additionally asks about campus food bank usage and provides an open-ended space for narrative context.

4.5.3. Expenses and housing security

The housing section measures housing security issues using a multi-select checklist, asks about landlord or building manager responses to prolonged issues, collects information on the current living situation and roommate context, and captures approximate monthly housing expenses for each individual. It also includes satisfaction and interest items related to subsidised graduate housing models. A top-three ranking question identifies the most significant housing problems, while an open-ended prompt provides further context.

4.5.4. Transportation expenses and commuting

Transportation measures include monthly transportation spending, one-way commute time, and a multi-select item on the main modes of transportation. The section also provides an open-ended prompt to capture circumstances not fully addressed by closed response options.

4.5.5. Workload and Finances

This module captures weekly hours spent on research-related activities and coursework, annual hours worked as a teaching or research assistant, and weekly hours worked in side employment outside assistantship roles. It also includes questions on whether paid work negatively impacts academic progress, the annual total funding package before tuition deductions, whether graduate funding covers daily living expenses, sources of additional funds when needed, and an estimate of additional annual compensation required to meet daily expenses. The module further includes a policy-oriented question on whether stipends should be tied to standardised metrics such as the Edmonton cost of living and a tuition sensitivity item describing the impact of a ten per cent tuition increase on basic needs.

4.6. Response Formats and How They Appear in the Dataset

The Google Forms export includes several response formats that affect the analysis workflow.

- Single-choice categorical items appear as text categories in the Comma-Separated Value file.
- Numeric and numeric-like entries include currency symbols, commas, and occasional ranges, all of which require consistent parsing for analysis.
- Multi-select items are stored as semicolon-separated selections within a single cell, requiring transformation into long format for frequency summaries.
- Ranking questions are represented by separate columns for rank 1, rank 2, and rank 3 selections, enabling both unweighted and weighted ranking summaries.
- Agree or disagree questions are stored as categorical responses, supporting proportion summaries and derived severity style indicators where appropriate.
- Open ended responses are stored as free text fields and are analysed using thematic summarisation approaches with careful privacy screening.

4.7. Consent, Anonymity, and Confidentiality Procedures

The survey introduction states that responses will be anonymous, that individual responses will remain confidential, and that results will be reported in brief form. The instrument also includes an explicit consent acknowledgement indicating that responses may be used anonymously in reporting and advocacy, including the use of written comments as non identifiable quotes.

Survey responses in the dataset used for analysis were anonymous, with one limited exception related to the prize draw. At the end of the survey, respondents who wished to enter the draw could voluntarily provide their University of Alberta email address. That email field was not required to complete the survey and was not relevant to the analytical objectives. To protect confidentiality, the email column was removed from the raw export prior to analysis and was not used in any reporting.

Open ended responses can contain indirectly identifying details even when no structured identifiers are collected. For this reason, any use of narrative comments in reporting has been preceded by screening and light redaction to remove personal names or uniquely identifying references, with preference given to aggregated thematic summaries.

4.8. Data Treatment and Preparation for Analysis

The CSV export uses full question text as column headers. Prior to analysis, responses were prepared to improve consistency while preserving respondents' intent. Preparation included standardising missing values, parsing numeric fields into consistent numeric form, harmonising minor label differences that could create duplicate categories, splitting multi-select responses to produce accurate frequency summaries, and constructing derived indicators for analysis, such as a food insecurity severity score and an essential cost index based on available monthly expense measures.

5. Analysis Approach

5.1 Overview and Analytical Goals

The analysis was designed to translate survey responses into a set of clear, policy-relevant indicators describing (1) typical graduate-student monthly costs, (2) the distribution of financial burden across respondents, and (3) how costs and hardship differ across key student groups. The approach prioritises interpretability plus advocacy usefulness over complex modelling. Results are presented primarily as descriptive summaries, subgroup comparisons, and a small set of derived indicators (e.g., indices and severity measures) that can be replicated in future waves of the survey.

5.2 Unit of Analysis and Reporting Strategy

The unit of analysis is the individual survey response. Findings are reported in two formats:

- Overall estimates: medians, interquartile ranges, and frequency distributions to describe "typical" costs and the spread of experiences.
- Stratified comparisons: metrics by subgroup where sample sizes permit (e.g., residency, enrolment, program, year, faculty, dependents). Results are reported with caution when cell sizes are small to avoid unstable or disclosive estimates.

5.3 Data Preparation and Variable Construction

Analyses use cleaned variables from the raw CSV export. Preparation steps include: converting inputs to numeric values, standardising labels (e.g., "Full time" vs "Full-time"), and splitting multi-select responses. Derived variables are defined explicitly:

5.3.1. Monthly expense variables
Key monthly expense measures include housing, food, and transportation, as well as other recurring monthly costs (such as utilities or healthcare). Numeric fields are parsed consistently (e.g., removing currency symbols and commas). Where responses are ranges, midpoints are used for plotting and summary statistics to retain information without exaggerating precision.

5.3.2. Essential Cost Index (ECI)
Essential Cost Index and housing share
To summarise overall monthly cost pressure, an Essential Cost Index is constructed from the three essential monthly expense items available consistently in the dataset: housing, food, and transportation. For respondent i, monthly housing cost is H_i, monthly food cost is F_i, and monthly transportation cost is T_i. The Essential Cost Index is defined as

$$ECL_i = H_i + F_i + T_i \tag{1}$$

A housing share measure is then computed when ECI_i is available and positive:

$$HS_i = \frac{H_i}{ECL_i} \tag{2}$$

Housing share captures the portion of essential monthly spending absorbed by housing. For regression interpretation, the Essential Cost Index is also expressed in thousands of dollars per month, and annual funding is expressed in thousands of dollars per year.

5.3.3. Food insecurity severity score

Food insecurity is analysed both as a direct indicator (yes/no, where present) and as a severity measure derived from the agree/disagree question. The severity measure equals the number of food-insecurity items the respondent marked “Agree” to. This provides a graded indicator that is more informative than a single binary indicator and supports comparisons across subgroups and cost-burden levels.

5.3.4. Ranking Questions (top three initiatives/challenges)

For ranking questions captured as three columns (Rank 1, Rank 2, Rank 3), results are summarised using:

- Any-rank frequency (how often an item appears anywhere in the top three), and
- a Borda-style score (3 points for Rank 1, 2 points for Rank 2, 1 point for Rank 3) to reflect priority intensity.

An exploratory regression model relating cost burden to food insecurity

To complement descriptive summaries, an exploratory logistic regression model is estimated to quantify the association between cost burden measures and the probability that a respondent reports any level of food insecurity. The purpose is to provide an interpretable summary of relationships that can support advocacy; it is not used to claim causal effects, particularly given the survey's voluntary nature.

Let $p_i = \Pr(Y_i = 1)$ denote the probability that the respondent i reports food insecurity. The model relates the log odds of food insecurity to basic monthly costs, housing share, and key circumstances available in the survey: annual funding $Fund_i$, international student status $Intl_i$, dependents Dep_i, part time enrolment PT_i, and weekly side job hours $SideHours_i$. The logistic regression specification is

$$log \frac{p_i}{(1-p_i)} = \beta_0 + \beta_1 ECI_i + \beta_2 HS_i + \beta_3 Fund_i + \beta_4 Intl_i + \beta_5 Dep_i + \beta_6 PT_i + \beta_7 SideHours_i \quad (3)$$

Predicted probabilities are computed using

$$p_i = \frac{\exp(\eta_i)}{1-\exp(\eta_i)} \quad (4)$$

$$\eta_i = \beta_0 + \beta_1 ECI_i + \beta_2 HS_i + \beta_3 Fund_i + \beta_4 Intl_i + \beta_5 Dep_i + \beta_6 PT_i + \beta_7 SideHours_i \quad (5)$$

Regression outputs are reported as coefficients and odds ratios with confidence intervals, and as predicted probability curves over the observed range of ECI_i, holding other covariates at typical values. For direct inclusion in the report, the model summary, odds ratio table, and prediction plots are exported as image files.

5.4 Descriptive Statistics and Visualisation Choices

Tables and figures are selected for lucidity and direct policy interpretation. Categorical variables with a small number of categories are summarised using frequency tables and simple charts. Longer categorical lists, multiple-select questions, and ranked outputs are summarised using compact count and percentage tables. Numeric variables such as monthly expenses, ECI_i, and commute time are summarised using distribution plots and percentile based summaries, with emphasis on medians and spread rather than means alone. Subgroup comparisons of costs and burden measures are visualised using distribution comparisons that show typical values and variability. To prevent visually misleading plots, very low count categories are grouped when appropriate, and overly granular categories are limited to the most common groups for plotting while full counts remain available in tables.

5.5 Subgroup Comparison Framework

Subgroup comparisons focus on dimensions most relevant to advocacy and planning, including residency status, dependents, enrolment status, degree program and year of study, and faculty where representation permits. For each subgroup, the analysis examines median monthly expenses by category, the distribution of ECI_i, housing share HS_i, and food insecurity measures, including both the binary indicator and the severity score. Subgroup tables present both counts and percentages, making denominators and reliability transparent.

5.6 Burden Thresholds and “High Strain” Flags

To support explicit communication of financial pressure, the analysis uses threshold based indicators that can be tracked over time. One indicator flags a high housing burden when housing share is at least 35 per cent of essential spending, $HS_i \geq 0.35$. A second indicator flags a high essential cost burden when ECI_i falls in the top quartile of the respondent distribution. These flags are reported overall and by subgroup to help identify groups most likely to experience high strain and to support targeted recommendations.

5.7 Handling Missing Data and Data Quality Limitations

Missing responses are expected in a voluntary survey. Analyses, therefore, use available-case calculations, meaning each table or figure includes only respondents with non-missing values for the variables required for that result. Denominators are reported alongside results so readers can see the effective sample size for each estimate. No imputation is performed to prevent the introduction of assumptions not supported by the raw responses. Extreme values are retained when plausible, but clearly implausible entries are treated as missing during numeric parsing. These choices preserve honesty and maintain a direct link between reported results and the raw CSV data.

5.8 Interpreting Results and Drawing Conclusions

Interpretation is intentionally conservative. The analysis shows where costs are highest, where hardship is most prevalent among respondents, and which groups appear disproportionately affected. Because the survey is not a random sample, findings are presented as evidence of the distribution of experiences among respondents and as credible indicators for advocacy, rather than as definitive population estimates. Results are translated into consequences for policy and support priorities, including which supports are most frequently prioritised, which cost categories dominate budgets, and how cost burden and funding adequacy relate to reported hardship.

6. Results

This section summarises the descriptive findings from the cost of living survey dataset (N = 1,026 submitted responses).

6.1 Sample characteristics

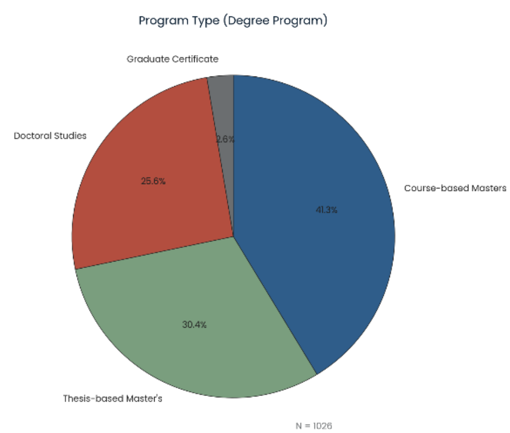

Figure 1. Graduate program type distribution.

Figure 1 shows that respondents were primarily enrolled in course based Masters programs (41.3%, n = 422), followed by thesis based Masters programs (30.5%, n = 312) and doctoral studies (25.6%, n = 262). Graduate certificate students represented a smaller share of the sample (2.6%, n = 27). This distribution conveys the survey captured perspectives across the most common graduate pathways, with a substantial representation of both research intensive and course based programs.

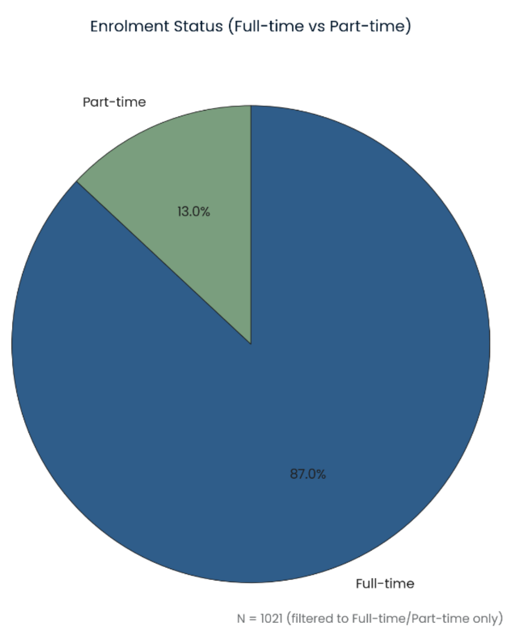

Figure 2. Enrolment status distribution.

Figure 2 summarises enrolment status. The sample is dominated by full time enrolment, with part time enrolment representing a smaller share.

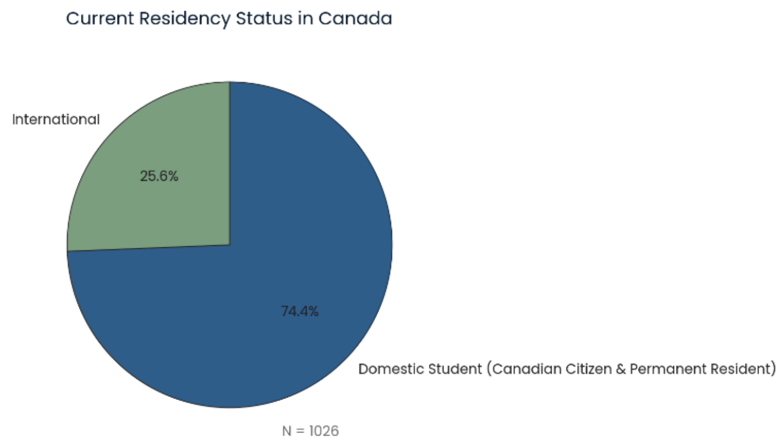

Figure 3. Residency status distribution.

Figure 3 shows that most respondents identified as domestic students (74.4%, n = 760), while international students comprised 25.6% (n = 263). This split is important for later comparisons because international students can face different fee structures, work limitations, and access to external supports.

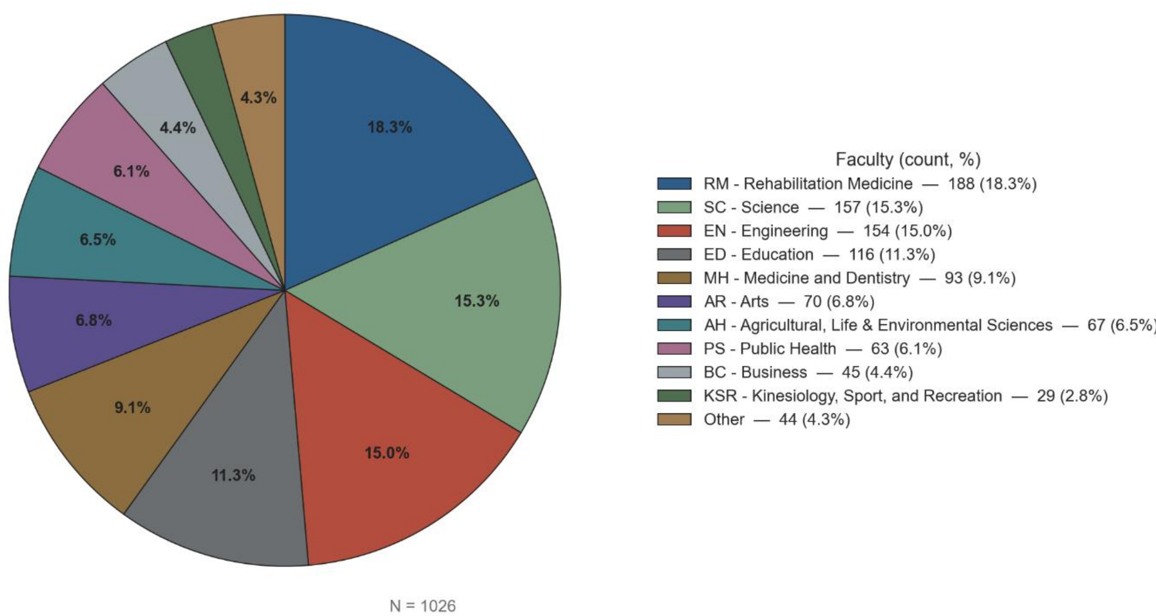

Figure 4. Faculty Representation.

Figure 4 indicates broad faculty coverage, with the largest shares from Rehabilitation Medicine (18.4%, n = 188), Science (15.2%, n = 156), Engineering (15.1%, n = 154), and Education

(11.1%, n = 114). Several other faculties contribute meaningful representation, supporting subgroup comparisons while still requiring care when interpreting very small faculty counts.

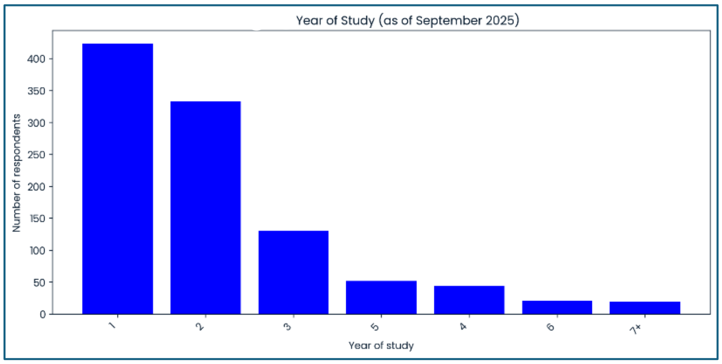

Figure 5. Year of study distribution.

Figure 5 shows that the sample skews toward earlier study years: Year 1 (41.3%, n = 423) and Year 2 (32.6%, n = 333) account for most responses. Years 3 and above make up the remainder. This pattern is consistent with the fact that monetary uncertainty and setup costs can be especially salient early in graduate study.

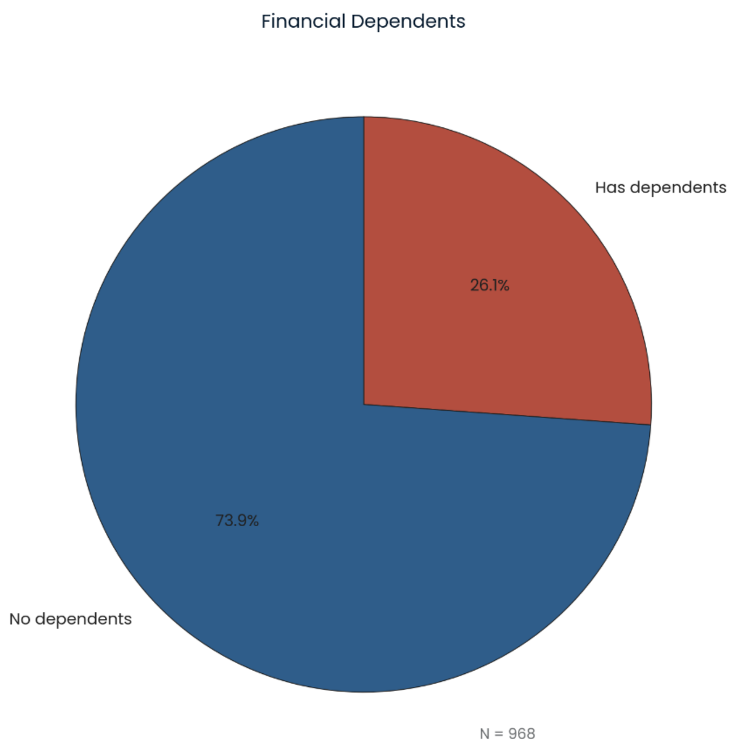

Figure 6. Dependent status.

Furthermore, Figure 6 shows that among respondents who answered the dependents question (n = 965), 26.1% (n = 252) reported supporting dependents financially, while 73.9% (n = 713) did not. This subgroup is conceptually important because caregiving responsibilities can shift both expenses and time available for paid work.

6.2 Food expenses and food insecurity

Monthly Food Expense

Food expense range	Count	Percent
$200–399	213	20.8%
$400–599	642	62.6%
$1000–1499	157	15.3%
$1500+	13	1.3%
Missing	1	0.1%

Table 1. Monthly food expense distribution.

Table 1 reports the distribution of self reported average monthly food expenses using the expense ranges available in the survey responses. The majority of respondents reported spending $400 to $599 per month on food (62.6%, n = 642). About one fifth reported spending $200 to $399 (20.8%, n = 213). Smaller shares reported spending $1000 to $1499 (15.3%, n = 157) and $1500 or more (1.3%, n = 13). Only one response was missing (0.1%, n = 1). Overall, the distribution is strongly concentrated in the $400 to $599 range, which provides a useful benchmark for typical monthly food costs among respondents.

Food Insecurity Indicators (Agree %)

Indicator (Agree %)	Description
53%	Could not afford balanced/nutritious meals
51%	Regularly relied on cheaper foods to avoid running out of money
23%	Worried food would run out before having money to buy more
17%	Skipped meals because of insufficient funds
13%	Food did not last and no money to buy more
3%	Went an entire day without eating due to lack of money

Figure 7. Food insecurity indicator summary.

Figure 7 summarises food insecurity experiences using the agree or disagree question. The highest agreement rates relate to restricted food quality and coping methods rather than extreme deprivation. Specifically, about half of respondents agreed that food costs prevented balanced or nutritious meals (53.0%, n = 542) and that they relied on a few low cost foods to avoid running out of money (51.2%, n = 524). Smaller but still meaningful shares agreed they worried that food would run out before they had money to buy more (22.5%, n = 230) or skipped meals due to insufficient funds (17.0%, n = 174). A smaller share (3.1%, n = 32) reported going an entire day without eating due to a lack of money. Taken together, these results suggest a pattern in which affordability pressures most often show up as reduced dietary quality and repeated coping behaviours, with a smaller subset reporting more serious outcomes.

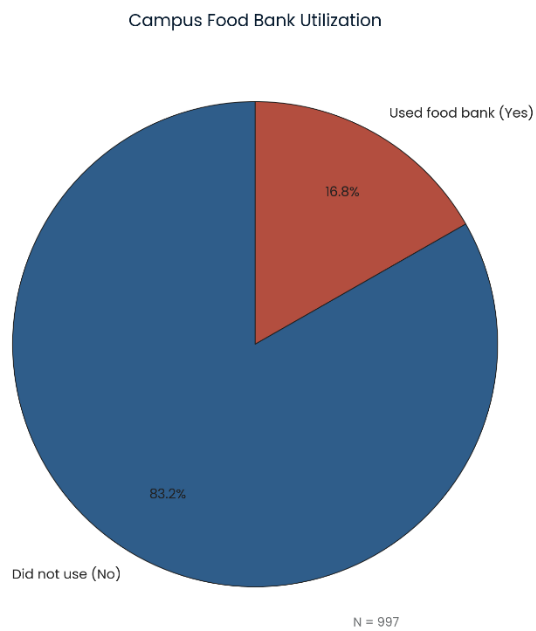

Figure 8. Campus Food Bank utilisation.

Figure 8 presents the reported utilisation of the University of Alberta Campus Food Bank during respondents' programs. This figure provides a direct indicator of reliance on emergency food support, complementing the wider insecurity measures above.

Food-Security Initiatives Ranked (N=1026 respondents answered at least one rank)

Initiative	Any-rank (Count)	Any-rank (%)	Rank 1	Rank 2	Rank 3	Borda (3-2-1)
Grocery Store Discounts (Loblaws, Metro, etc.)	878	**85.6%**	566	234	130	**2296**
Subsidized Graduate Student Grocery Store	746	**72.7%**	190	298	273	**1439**
Subsidized Meal Kit Delivery Services (Hellofresh, ChefsPlate, etc.)	538	**52.4%**	81	271	198	**983**
Campus Food Bank	378	**36.8%**	145	96	173	**800**
Communal Kitchen with Subsidized Pre-made Food	191	**18.6%**	27	67	100	**315**
Communal Kitchen with Subsidized Ingredients	113	**11.0%**	12	25	78	**164**
Communal Fridges with Subsidized Food Options	114	**11.1%**	5	35	74	**159**

Table 2. Ranked food security initiatives.

Table 2 summarises the ranked food security initiatives using an aggregate scoring approach. Items placed frequently in the top ranks represent the clearest priorities for program design and advocacy, because they reflect both the prevalence and the intensity of preference (not just a single first choice).

6.3 Housing costs and housing security

Monthly Housing Expense — Distribution

Monthly housing expense range	Count	Percent
$0–799	**182**	**17.7%**
$800–1199	**280**	**27.3%**
$1200–1599	**245**	**23.9%**
$1600–1999	**277**	**27.0%**
Missing	**42**	**4.1%**

Table 3. Monthly housing expense distribution.

Table 3 summarizes self reported monthly housing expenses using the expense ranges shown in the survey distribution table. The most common categories were $800 to $1199 (27.3%, n = 280) and $1600 to $1999 (27.0%, n = 277), followed by $1200 to $1599 (23.9%, n = 245) and $0 to $799 (17.7%, n = 182). A total of 42 responses were missing (4.1%). Overall, the distribution indicates that a large share of respondents reports housing expenses above $800 per month, with a substantial proportion concentrated in the $1,600 to $1,999 range, suggesting significant housing pressure for many students before other essentials are considered. The housing expense question asks for individual monthly housing expenses and explicitly includes rent or mortgage, as well as maintenance fees and utilities, so respondents in the highest range may be reporting a larger unit, fewer opportunities to split costs with roommates, or a wider definition of housing costs that includes utilities. In the survey data, the highest housing expense group is much more likely to include students supporting dependents: about half of respondents in the top housing category report supporting dependents financially (133 of 263, 50.6%) (263 is the number of those respondents who also answered the dependents question), compared to a much smaller share in the lowest housing category (22 of 176, 12.5%). The living situation responses are consistent with this pattern: the highest housing group reports living with a significant other or family more often, while the lowest housing group is more commonly associated with roommate living.

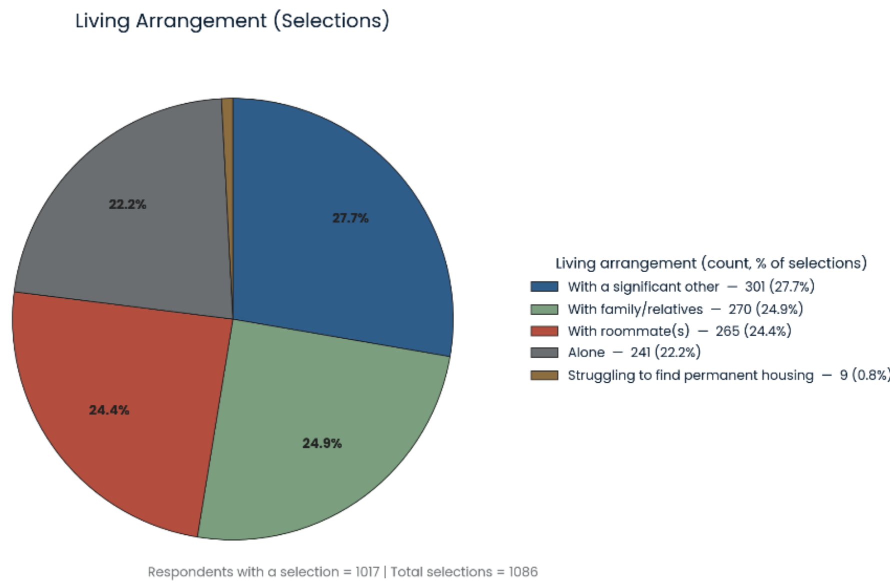

Figure 9. Living situation and roommates.

Figure 9 describes respondents' living situations (for example, living alone, with roommates, with family, or with a partner). This figure contextualises the housing cost table because household structure is a main driver of rent sharing and total monthly burden. Where applicable, roommate counts help interpret why some respondents fall into lower or higher housing cost brackets.

Housing Insecurity Issues (Top 15 by Selections)

Housing issue	Count	Percent
No	603	58.3%
Lacks proper heating/cooling	111	10.7%
Cannot afford housing	79	7.6%
Improper sanitation, i.e., mold growth, humidity issues	76	7.4%
Pest infestation	58	5.6%
Overcrowding	43	4.2%
Living transiently in various locations	23	2.2%
No challenges personally	1	0.1%
I have adequate housing, though it is expensive for what I am currently making	1	0.1%
not now, but previously I have.	1	0.1%
Living beyond my means	1	0.1%
Rent being raised $360 in one year at my longterm 1 bedroom apartment rental forced me to move a few months ago to the outskirt of the city	1	0.1%
Living at home with parents because I cannot afford to move out	1	0.1%
Can not afford critical home repairs	1	0.1%
rental is expensive	1	0.1%

Table 4. Housing Insecurity Issues.

Table 4 summarises reported housing insecurity issues based on selections from the housing security checklist. The most common response was No (58.3%, n = 603), indicating that a majority of selections reflected no housing insecurity issue. Among respondents who reported an issue, the most frequently selected concerns were a lack of proper heating or cooling (10.7%, n = 111), cannot afford housing (7.6%, n = 79), and improper sanitation, such as mould growth or humidity issues (7.4%, n = 76). Other recurring issues included pest infestation (5.6%, n = 58) and overcrowding (4.2%, n = 43), while living transiently in various locations was less common (2.2%, n = 23). A small number of additional responses appeared as one-off entries (0.1% each), reflecting individualised circumstances. A total of 79 students reported being unable to afford housing, a figure that elicits serious concern. Notably, this group is disproportionately composed of international students. Approximately half of those indicating housing unaffordability are international students (40 of 79, 50.6%), despite international students representing a much smaller proportion of the overall sample. This difference is significant because the survey's annual funding measure is reported before tuition is deducted, and tuition obligations can substantially reduce the funds available for rent and daily expenses. Furthermore, while international students are required to demonstrate proof of funds upon arrival, this requirement does not ensure ongoing affordability after accounting for tuition, deposits, moving costs, and recurring rent payments.

The issue is mainly linked to funding adequacy rather than to housing conditions. Among respondents who reported being unable to afford housing and answered the funding adequacy question, the majority indicated that they could not cover daily expenses exclusively through graduate funding (58 of 69, 84.1%). This evidence suggests that the inability to afford housing is closely associated with an overall budget shortfall, rather than simple dissatisfaction with housing quality or conditions.

Respondents who reported unaffordable housing frequently rely on financial resources beyond their graduate funding. Among those who answered the question regarding additional support sources (n = 72), the most common strategies included using personal savings (40 of 72, 55.6%), relying on family or relatives (34 of 72, 47.2%), taking loans or lines of credit (29 of 72, 40.3%), and seeking external employment such as a second job (24 of 72, 33.3%). These patterns signify that unaffordable housing is often managed through temporary financial buffers and debt rather than through stable income growth.

Responses about living situations reveal the practical strategies students use to manage housing costs. Many respondents in this group report living with family or relatives (31 of 79, 39.2%) or with roommates (25 of 79, 31.6%), reflecting efforts to reduce expenses using shared housing. A considerable minority (8 of 79, 10.1%) reports difficulty securing permanent housing, implying a more acute concern about housing stability for some students.

It is essential to interpret the "PhD funding minimum" in terms of monthly budgeting. For example, a typical annual funding package of $25,000 before tuition amounts to approximately $2,083 per month before any deductions. If housing costs are at the higher end of the ranges reported in the survey, housing expenses alone may consume most of this monthly amount, leaving insufficient funds for food, transportation, and other necessities. This context clarifies why students may report housing unaffordability despite receiving graduate funding, especially when funding is calculated before tuition and other necessary expenses are considered. Overall, the distribution suggests that although many respondents did not report housing insecurity through this checklist, a substantial minority identified housing quality and affordability issues, with heating or cooling and housing affordability emerging as the most prominent concerns.

Top Housing Challenges (Ranked) — N=1026 answered at least one rank (Top 15)

Challenge	Any-rank (Count)	Any-rank (%)	Rank 1	Rank 2	Rank 3	Borda (3-2-1)
Affordibility	**964**	**94.0%**	734	117	113	**2549**
Proximity to campus	**829**	**80.8%**	152	437	240	**1570**
Safe and secure	**568**	**55.4%**	62	217	289	**909**
Accessible ETS routes	**464**	**45.2%**	50	154	260	**718**
Accessible housing	**164**	**16.0%**	12	69	83	**257**
Accessible childcare	**89**	**8.7%**	16	32	41	**153**

Figure 10. Housing insecurity and housing difficulties ranking.

Figure 10 summarises the reported housing security issues and the ranked challenges students face (e.g., affordability, proximity to campus, safety, and availability). The ranking results help move beyond costs alone by identifying which aspects of housing are most disruptive or limiting for graduate students.

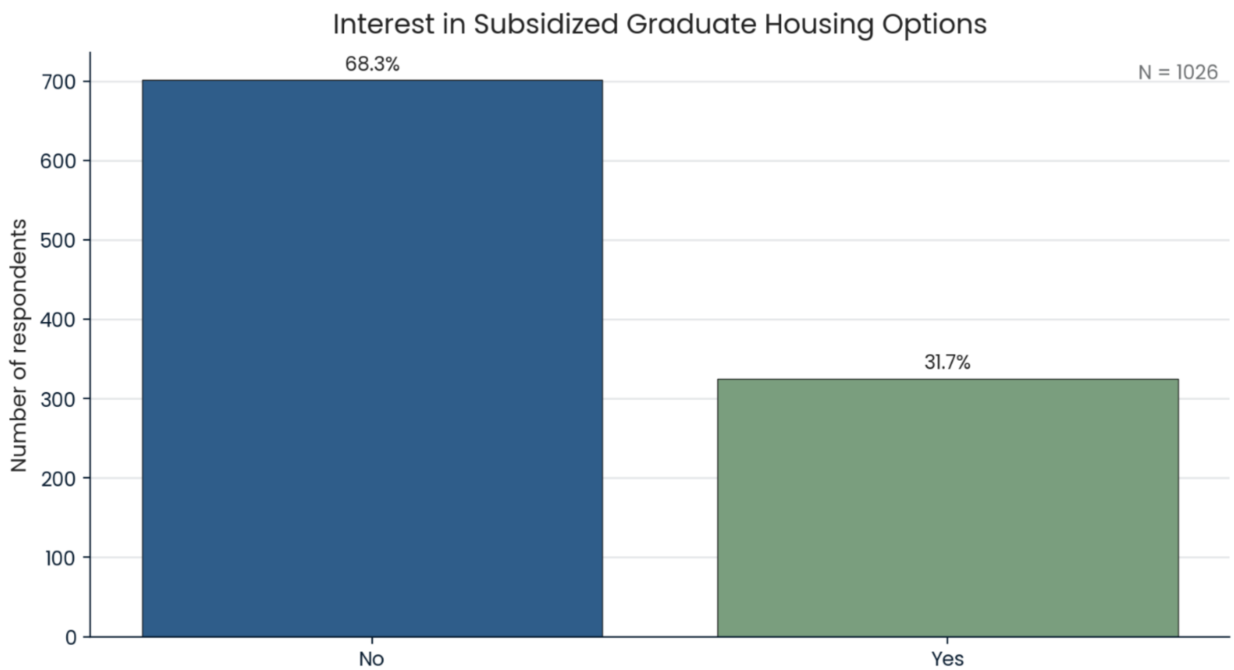

Figure 11. Interest in subsidised graduate housing options.

Figure 11 reports interest in subsidised dorm-style graduate housing configurations. This is a policy-relevant indicator because it measures potential uptake and demand for specific housing interventions rather than documenting only hardship.

6.4 Transportation costs and access

Monthly Transportation Expense — Distribution

Monthly transport expense range	Count	Percent
$0–49	230	22.4%
$50–99	141	13.7%
$100–149	177	17.3%
$150–199	74	7.2%
$200–299	149	14.5%
$300–499	150	14.6%
$500+	105	10.2%

Table 4. Monthly transportation expense distribution.

Table 4 summarises reported monthly transportation spending. The distribution is wide, with many respondents clustered in lower monthly spending values and a smaller subset reporting high transportation spending. This variation is expected because transportation costs depend heavily on commute distance, transport mode, and car related fixed costs.

Transport Modes (Normalized Categories)

Mode (normalized)	Count	Percent
Public transit	653	36.0%
Drive	509	28.0%
Walk	471	26.0%
Bike	155	8.5%
Ride-hailing / taxi	15	0.8%
Carpool / rideshare	4	0.2%
Scooter	3	0.2%
Skates	2	0.1%
Car	1	0.1%
Not in Edmonton / remote	1	0.1%
Car Share	1	0.1%

Table 5. Main modes of transportation (normalised).

Table 5 shows the most common transportation modes after normalising duplicate text responses. Public transit was most frequently selected (36.0%, n = 653 selections), followed by driving (28.0%, n = 509), walking (26.0%, n = 471), and biking (8.5%, n = 155). Rare responses such as rideshare variants were grouped to avoid duplicate rows and improve interpretability. The results show that transit and active transportation together account for a large share of commuting behaviour, carrying consequences for the value of transit pass pricing, service reliability, and campus accessibility.

Commute Time Distribution (N=1026)

Commute time (Minutes)	Count	Percent
5–15	240	23.4%
15–30	396	38.6%
30–45	232	22.6%
45–60	117	11.4%
60+	41	4.0%

Table 6. Commute time distribution.

Table 6 summarises the one-way commute time. The largest group reported 15 to 30 minutes (38.7%, n = 396), followed by 5 to 15 minutes (23.4%, n = 240), 30 to 45 minutes (22.6%, n = 232), 45 to 60 minutes (11.4%, n = 117), and 60 minutes or more (4.0%, n = 41). This distribution suggests that most respondents commute within 45 minutes one way, but a nontrivial minority experiences long commutes that can compound time pressure and transportation costs.

6.5 Workload and finances

Weekly Hours on Courses/Classes — Distribution

Weekly hours range	Count	Percent
10–14	47	4.6%
15–19	140	13.6%
20–29	140	13.6%
30–39	199	19.4%
40–59	250	24.4%
60+	59	5.8%
Missing	191	18.6%

Table 7. Weekly hours on research and coursework.

Table 7 summarises weekly workload hours. This table is included because time constraints interact with monetary constraints: students facing high costs may seek additional paid work, but workload demands can limit capacity to do so.

Paid work exposure summary (cleaned for implausible entries)

Measure	N	Missing	Mean	Median	P25	P75	P90	Max
TA or RA hours per year (cleaned)	828	198	122.14	0.00	0.00	32.75	384.00	2400.00
Side job hours per week (cleaned)	864	162	9.40	0.00	0.00	15.00	35.00	60.00
TA or RA weekly equivalent (annual divided by 52, cleaned)	828	198	2.35	0.00	0.00	0.63	7.38	46.15
Total paid hours per week (weekly equivalent plus side job, cleaned)	800	226	11.42	5.23	0.00	17.54	38.46	76.92

Table 8. Teaching and research assistant hours and side job hours.

Table 8 describes paid work exposure through reported TA or RA hours and side job hours. This provides context to understand economic stress measures, as employment intensity can both react to financial pressure and contribute to educational stress.

Workload — Impact of Working on Academic Progress

Work impact response	Count	Percent
No	443	61.6%
Yes	276	38.4%

Table 9. Perceived academic impact of working.

Table 9 summarises respondents' views on whether TA, RA, or side job work negatively impacts academic progress. This connects the cost of living problem to academic outcomes, supporting advocacy arguments that monetary stress is not only a welfare issue as well as an academic success issue.

Primary Source of Financial Support (Normalized) — Frequency Table

Response (normalized)	Count	Percent
Graduate assistantship stipend (TA/RA)	500	48.7%
Student loans	208	20.3%
Personal savings	123	12.0%
Family support	76	7.4%
Part-time employment	61	5.9%
Full-time employment	31	3.0%
Other (rare responses)	22	2.1%
Tuition remission	5	0.5%

Table 10. Primary financial support source.

Table 10 reports the most common primary support sources. The leading category was graduate assistantship stipend (32.5%, n = 333), followed by student loans (20.1%, n = 206) and scholarships or grants (15.7%, n = 161). Personal savings (11.7%, n = 120) and family support (7.4%, n = 76) were also common. Smaller shares reported part time employment (5.9%, n = 61) or full time employment (1.5%, n = 15). This composition matters because reliance on loans and savings denotes potential long term financial risk, while reliance on assistantships signals sensitivity to stipend levels and contract availability.

Finances — Sources Used When Additional Money is Needed (Normalized)

Source (normalized)	Count	Percent
Personal savings	512	26.6%
Family or relatives	435	22.6%
External employment (e.g., second job)	348	18.1%
Loans (student loans, line of credit, etc.)	337	17.5%
TA/RA/Fellowship	188	9.8%
No additional support	88	4.6%
Other (rare responses)	10	0.5%
Partner/spouse support	9	0.5%

Table 11. Funding adequacy and additional support sources.

Table 11 summarises whether graduate funding alone covers day to day expenses and the sources used when additional money is needed. These outputs directly support advocacy by translating “financial hardship” into concrete behaviours, such as drawing down savings, taking loans, or relying on family.

Finances — Additional Compensation Needed (Summary)

Metric	Value
N (non-missing)	**599**
Missing (count)	**427**
Missing (%)	**41.6%**
Mean	**$10,904**
Median	**$7,000**
P25	**$3,000**
P75	**$15,000**
P90	**$25,000**
Max	**$80,000**

Table 12. Additional compensation is needed.

Table 12 summarises the distribution of the additional annual compensation respondents report needing to meet day to day expenses. This figure provides a direct, interpretable policy quantity that can be discussed alongside stipend levels and cost benchmarks.

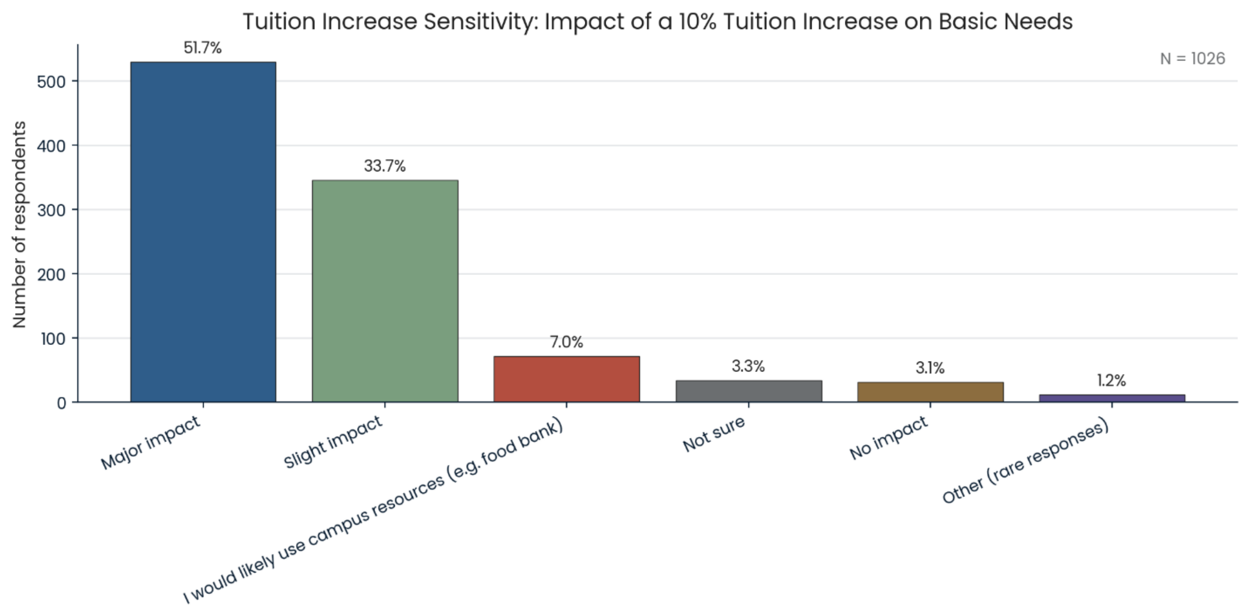

Figure 12. Tuition increases sensitivity.

Figure 12 shows how a 10% tuition increase would affect basic needs. This result delivers a policy stress test: it identifies how close to the margin many students already are, and it frames tuition changes in terms of basic needs impacts rather than abstract affordability.

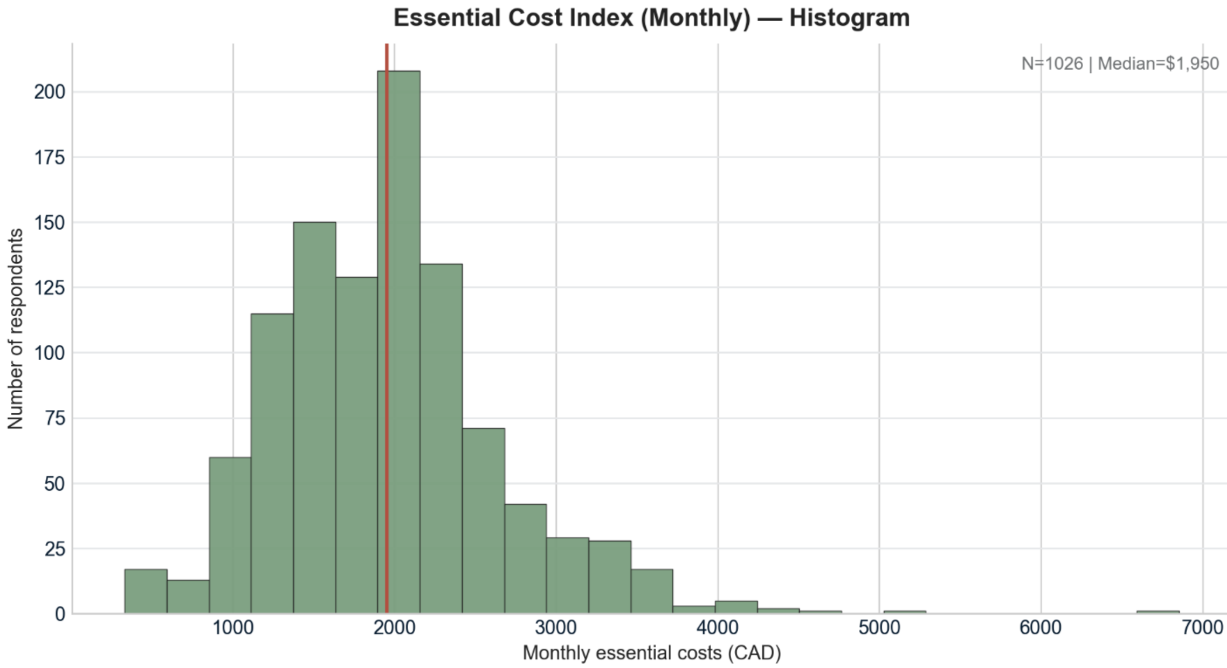

Figure 13. Essential Cost Index (Monthly) - Histogram.

Figure 13 presents the distribution of the Essential Cost Index (monthly), constructed as the sum of three self reported expense measures: individual monthly housing expenses (rent or mortgage and related costs such as maintenance fees and utilities), average monthly food expense, and monthly transportation spending. Across respondents (N = 1,026), the distribution is right skewed, indicating that while most students cluster in a mid range of essential monthly costs, a smaller subset faces substantially higher combined expenses. The median Essential Cost Index is $1,950 per month, as indicated by the vertical reference line, suggesting that a typical respondent spends roughly $2,000 per month on these core necessities alone. The long upper tail, extending beyond $4,000 and reaching approximately $7,000 in rare cases, highlights that a nontrivial minority experiences markedly higher essential cost burdens, which can substantially reduce financial flexibility even before accounting for other recurring costs such as tuition, healthcare, or miscellaneous academic expenses.

```
                           Logit Regression Results
==============================================================================
Dep. Variable:        Y_food_insecure   No. Observations:                  983
Model:                          Logit   Df Residuals:                      975
Method:                           MLE   Df Model:                            7
Date:                Mon, 09 Mar 2026   Pseudo R-squ.:                 0.07422
Time:                        20:39:11   Log-Likelihood:                -629.84
converged:                       True   LL-Null:                       -680.33
Covariance Type:            nonrobust   LLR p-value:                 6.731e-19
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const         -0.3235      0.545     -0.593      0.553      -1.392       0.745
ECI_k         -0.3119      0.128     -2.432      0.015      -0.563      -0.061
HS             1.7503      0.681      2.572      0.010       0.416       3.084
Fund_k        -0.0185      0.006     -2.962      0.003      -0.031      -0.006
Intl           0.9529      0.165      5.789      0.000       0.630       1.276
Dep            0.1691      0.175      0.965      0.335      -0.174       0.513
PT            -0.7951      0.237     -3.350      0.001      -1.260      -0.330
SideHours      0.0005      0.002      0.296      0.767      -0.003       0.004
==============================================================================
```

Figure 14. Regression Results.

The logistic regression model predicting whether a respondent reports any level of food insecurity used 983 observations and converged successfully. Overall model fit indicates that the included predictors jointly improve explanation of food insecurity relative to a null model, with a likelihood ratio test p value of $6.73e^{-19}$ and a McFadden pseudo R squared of 0.074. Several predictors show statistically significant associations and help explain the pathways through which financial strain likely operates.

First, housing share is positively associated with food insecurity (coef = 1.750, p = 0.010), meaning that respondents who devote a larger portion of their essential spending to housing have higher odds of reporting food insecurity. This result is consistent with a budget trade off mechanism: when housing absorbs a larger share of the basic spending basket, there is less flexibility to adjust other essentials, and food becomes the category most likely to be reduced. Importantly, this finding is about the composition of essential spending, not only about absolute housing costs. It helps explain why some respondents can appear "high spending" overall and still experience food insecurity if a large fraction of their essential spending is tied up in housing.

Second, international student status is strongly associated with higher food insecurity (coef = 0.953, p < 0.001). The survey cannot identify a single causal reason for this gap, but several mechanisms are plausible and consistent with how the funding variables are measured. Funding in the survey is reported before tuition is deducted, so net disposable resources can differ substantially across groups even when gross annual funding appears similar. International students also face different fee structures and may experience additional upfront costs or constraints that are not fully captured in the monthly essentials index. Therefore, the international coefficient should be interpreted as capturing a systematic difference in vulnerability within the respondent sample after controlling for the included cost and funding measures, rather than as evidence of a single specific cause.

Third, higher annual funding is associated with lower food insecurity (coef = −0.0185 per additional $1,000, p = 0.003), which aligns with the basic expectation that increased financial resources reduce hardship. The model also shows a negative association for part time enrolment (coef = −0.795, p = 0.001). This does not necessarily mean part time status protects against hardship. It may reflect that the part time subgroup includes respondents with additional household income sources or different living arrangements that are not fully captured by the included variables. In other words, enrolment status may be acting as a proxy for unmeasured differences in economic context.

Finally, the Essential Cost Index is negatively associated with food insecurity in this specification (coef = −0.312, p = 0.015). This result can appear counterintuitive at first, but it is consistent with how the model controls for both funding and housing share simultaneously. Conditional on funding levels and the share of essentials devoted to housing, higher reported essential spending may partially reflect greater overall resources or higher capacity to spend, rather than greater deprivation. This highlights why the share measure is useful: food insecurity is more strongly linked to budget constraint and spending composition than to spending levels alone.

Dependent status and weekly side job hours are not statistically distinguishable from zero in this model (p = 0.335 and p = 0.767). This does not imply these factors are unimportant in reality. It indicates that, within this respondent sample and with the current set of controls, their independent association is not estimated precisely enough to separate from zero.

Odds ratios with 95 percent CI

	coef	odds_ratio	ci_low_or	ci_high_or
const	-0.3235	0.7236	0.2485	2.1071
ECI_k	-0.3119	0.7321	0.5694	0.9413
HS	1.7503	5.7565	1.5163	21.8539
Fund_k	-0.0185	0.9817	0.9698	0.9938
Intl	0.9529	2.5933	1.8782	3.5807
Dep	0.1691	1.1842	0.8399	1.6697
PT	-0.7951	0.4515	0.2835	0.719
SideHours	0.0005	1.0005	0.9973	1.0037

Table 13. Odds Ratios.

Table 13 reports odds ratios from the logistic regression predicting whether a respondent reports any level of food insecurity. Holding other variables constant, a higher housing share is strongly associated with higher odds of food insecurity, with an odds ratio of 5.76 and a 95 per cent confidence interval from 1.52 to 21.85. Because housing share is a proportion, a more understandable change is 0.10, corresponding to a 10 percentage-point increase and about 1.19 times the odds of food insecurity. Annual funding is associated with lower odds of food insecurity, with an odds ratio of 0.98 per additional $1,000 in funding and a 95 per cent confidence interval from 0.97 to 0.99. International students have substantially higher odds of food insecurity than domestic students, with an odds ratio of 2.59 and a 95 per cent confidence interval from 1.88 to 3.58. Part-time enrolment is associated with lower odds of food insecurity, with an odds ratio of 0.45 and a 95 per cent confidence interval from 0.28 to 0.72. The Essential Cost Index, measured in thousands of dollars per month, is also negatively associated with food insecurity in this specification, with an odds ratio of 0.73 and a 95 per cent confidence interval from 0.57 to 0.94, which likely reflects that spending levels can capture both cost exposure and ability to pay once housing share and funding are included. Dependent status and weekly side job hours are not statistically distinguishable from zero in this model because their confidence intervals include 1.

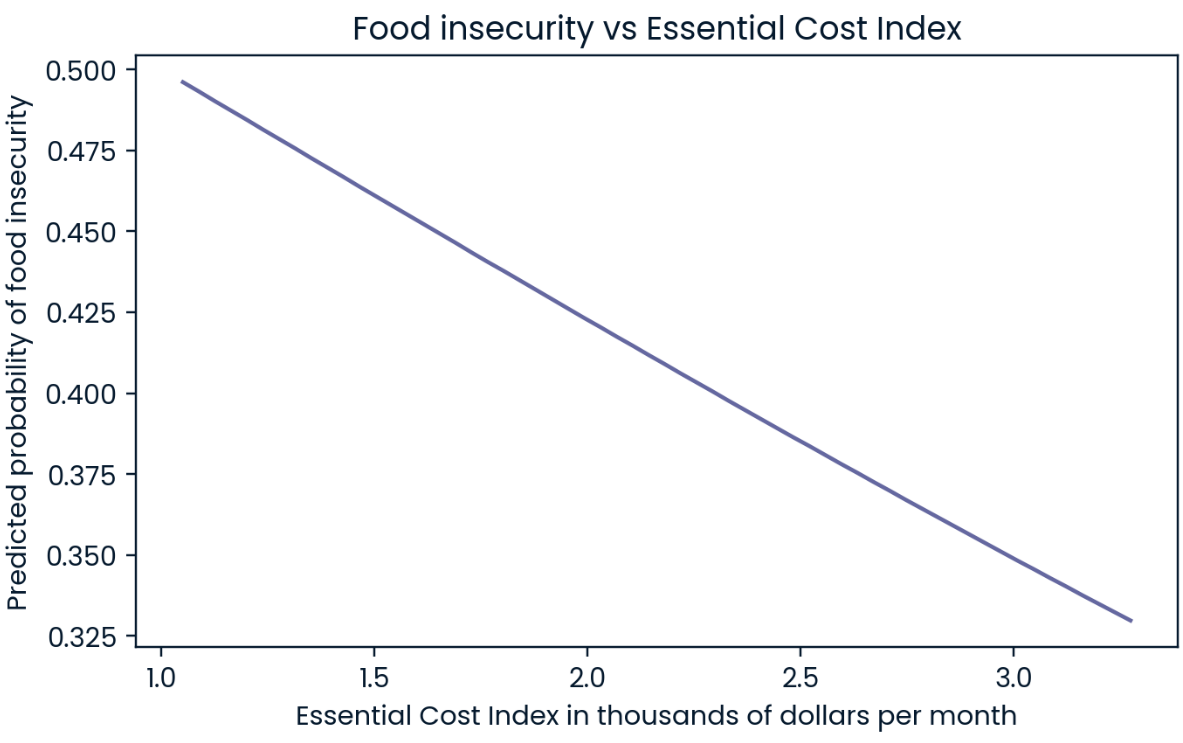

Figure 15. Food insecurity vs Essential Cost Index.

Figure 15 plots predicted probabilities of reporting any level of food insecurity across the observed range of the Essential Cost Index, holding other model variables at typical values. The predicted probability declines as the Essential Cost Index increases, falling from about 0.50 at roughly 1.0 thousand dollars per month to about 0.33 at roughly 3.3 thousand dollars per month. This pattern should be interpreted cautiously because the model includes both the total level of essential spending and its composition, measured by housing share, as well as annual funding. Conditional on housing share and funding, higher total essential spending can reflect greater overall resources or different living circumstances rather than greater monetary stress alone. Within this specification, the figure suggests that the strongest signal of cost-related strain is not the absolute level of essential spending alone, but how it is allocated toward housing and the structural differences captured by variables such as international status and funding.

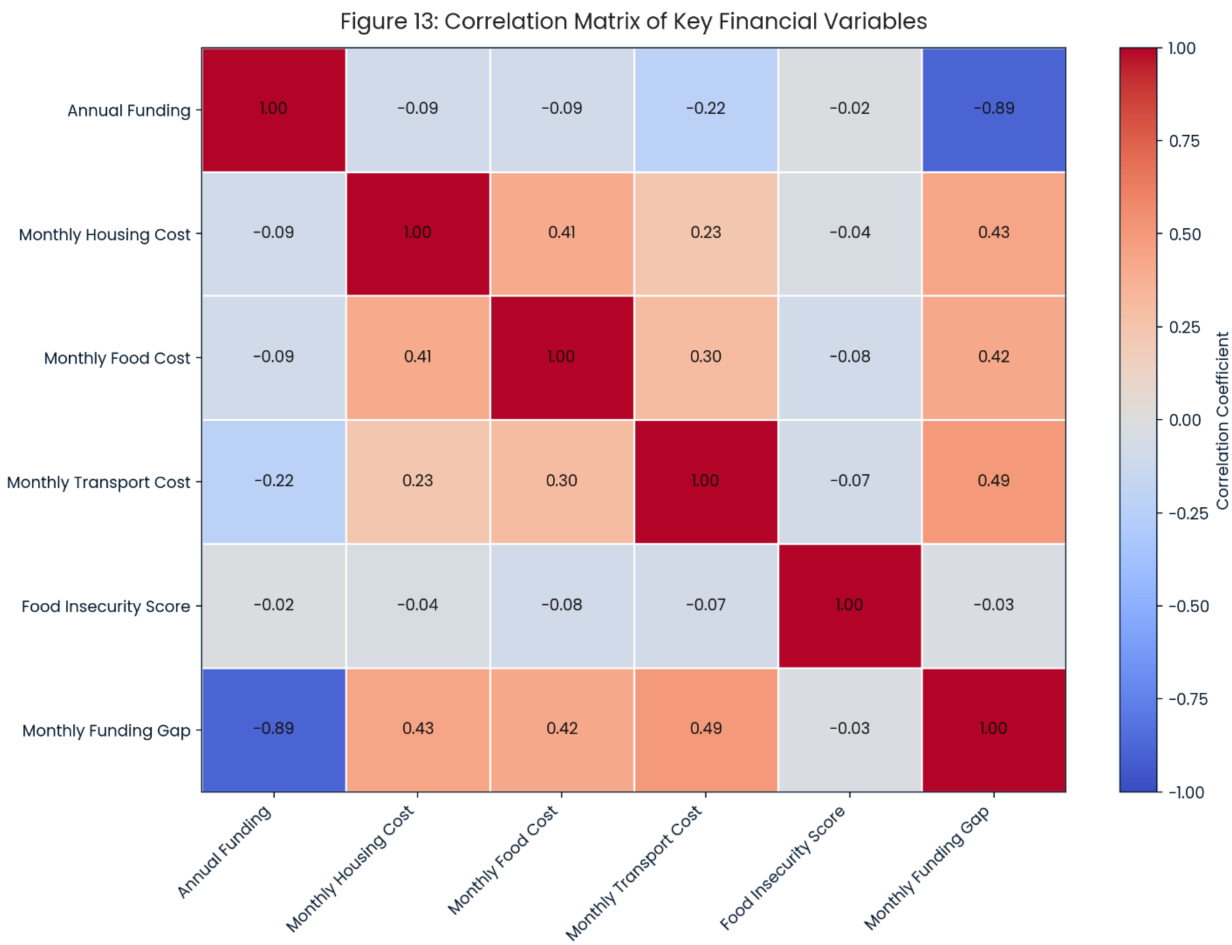

Figure 16. Correlation Matrix of Key Financial Variables.

Figure 16 summarises pairwise correlations among key financial measures, including annual funding, monthly essential cost components, a food insecurity severity score, and the monthly funding gap.

The three monthly cost measures tend to rise together rather than move in opposite directions. Housing cost has a moderate positive relationship with food cost ($r = 0.41$) and a smaller positive relationship with transportation cost ($r = 0.23$). Food and transportation costs are also positively related ($r = 0.30$). In practical terms, students who report higher housing expenses also tend to report higher food and transportation spending, suggesting that cost pressures often occur simultaneously across multiple essentials rather than being offset by savings in another category.

Second, annual funding shows weak relationships with the cost components. Correlations between annual funding and monthly housing cost ($r = -0.09$) and monthly food cost ($r = -0.09$) are close to zero, while the relationship with monthly transport cost is slightly more negative ($r = -0.22$). This pattern indicates that higher annual funding is not strongly aligned with higher monthly costs in this respondent sample. In other words, cost exposure appears only weakly compensated by funding levels.

Third, the monthly funding gap behaves as expected by construction. The funding gap is strongly negatively correlated with annual funding ($r = -0.89$), reflecting that higher funding reduces the gap. At the same time, the funding gap is moderately positively correlated with monthly housing ($r = 0.43$), food ($r = 0.42$), and transport ($r = 0.49$) costs, indicating that increases in any essential cost category are associated with a larger shortfall between monthly costs and monthly funding.

Finally, the food insecurity score has near-zero correlations with the financial variables ($r = -0.08$ to -0.02 with funding and costs, and $r = -0.03$ with the funding gap). This does not imply that costs are unrelated to food insecurity in practice. It indicates that within this sample and this bivariate correlation framework, the measured food insecurity severity does not track linearly with reported costs or funding. Food insecurity likely depends on additional factors not captured in this correlation matrix, such as household size, dependents, debt obligations, unanticipated expenses, timing of income, or access to external support.

Overall, Figure 13 suggests that essential costs tend to rise together and that the funding gap is most sensitive to both funding levels and transportation and housing costs. The weak relationship between annual funding and the cost components supports the policy interpretation that funding does not automatically scale with higher cost exposure, which reinforces the argument for cost aligned funding benchmarks and targeted supports.

6.6 Summary of key patterns

Across the figures and tables, three reliable patterns emerge. First, essential costs, particularly housing and food, appear considerable for a large share of students and are distributed unevenly across the sample. Second, food insecurity is not limited to rare extreme cases; indicators of reduced food quality and reliance on coping strategies are widespread. Third, financial pressure is closely linked to workload and support structures, with many students relying on assistantships, loans, or savings and reporting that paid work can affect academic progress. These descriptive findings supply the empirical base for the interpretation and recommendations that follow.

7. Interpretation and Discussion

7.1 What the results show about graduate-student cost pressure

Together, the survey results show that essential monthly expenses are both high and uneven among respondents. Housing and food are the biggest expenses, creating recurring obligations that many students must meet with limited, often fixed incomes (such as stipends, scholarships, and assistantships). The Essential Cost Index and related charts highlight two key patterns for advocates:

- Median values show the typical student experience, but the right tail reveals some face much higher essential costs. These higher costs matter most for policy because they identify students at greater risk of hardship and leaving their program.
- Cost pressure is not evenly distributed. When subgroup sizes allow, some groups and life situations face higher costs. This matters because targeting these groups may reduce risk more than broad inflation-based messaging.

7.2 Burden and affordability: why the "share" indicators matter

A significant contribution of this analysis is the transition from reporting individual expense categories in isolation to employing burden metrics that facilitate interpretation in policy contexts. Specifically, measures such as housing share and high burden threshold flags consolidate numerous cost categories into a more coherent assessment of affordability pressure and monetary vulnerability.

Housing share is defined as monthly housing expenses divided by the Essential Cost Index, where the Essential Cost Index represents the sum of monthly housing, food, and transportation costs reported in the survey. This indicator reflects the allocation of a respondent's essential spending and the prominence of housing within their overall cost structure. It does not assess affordability relative to total income, as the survey-based index excludes comprehensive income data and certain expense categories. Therefore, these ratios should be interpreted as survey-based burden indicators rather than comprehensive income-based affordability ratios.

Despite this limitation, the indicators deliver substantial utility. A high housing share indicates that a significant portion of measured essential costs is allocated to housing, reducing flexibility in other essential categories such as food and transportation. The threshold flags help identify students with potentially fragile budgets who are more susceptible to financial shocks, including rent increases, higher utility costs, tuition increases, or unexpected health-related expenses. Additionally, these measures are repeatable, enabling the GSA and the University to monitor changes in burden levels over time and to evaluate the effectiveness of governmental interventions in reducing the proportion of students experiencing high cost pressure.

7.3 Food insecurity as both a cost and a lived-experience outcome

The food insecurity findings strengthen the report because they connect financial measures to results that reflect daily wellbeing. The agree/disagree question captures experiences such as worry about running out of food, reduced diet quality, reliance on low-cost foods, and skipping meals. Summarising these responses into a severity score adds interpretive value by showing that food insecurity exists on a spectrum rather than as a binary condition.

From a policy perspective, two points are especially salient:

- Food insecurity signals that financial resources are inadequate. Even if food spending looks reasonable, insecurity means students are cutting back, facing unpredictability, or shifting money to other essentials.
- Support usage does not necessarily match need. If food bank utilisation is lower than the prevalence of insecurity, this suggests barriers such as stigma, awareness, eligibility uncertainty, location/hours constraints, or the mismatch between the service model and student schedules. This gap can guide recommendations to expand support and improve access.

7.4 Housing and transportation constraints: structural, not individual choices

Housing and transportation patterns should be seen as systemic limitations, not simply individual budgeting choices. Key takeaway: For many students, costs and commutes continue to be determined by local rental markets, availability, and transit limitations, making these factors largely outside individual control.

When analysing commute time, transport mode, and spending, it shows that transportation is often a hidden cost. Students farther from campus may save on rent but face longer commuting and sometimes elevated transportation costs. This trade-off matters regarding policies on transit subsidies, campus housing, and support for both financial and time poverty.

7.5 Workload, employment, and the trade-off between income and academic progress

Workload analysis clarifies cost data with graduate student time constraints. Weekly hours on research and coursework show expected educational obligations. Adding paid work and funding adequacy highlights a trade-off:

- Students who must secure additional income through paid work may face constraints on research productivity, time-to-degree, and wellbeing.
- Funding that falls short is not simply a budgeting issue; it forces students to choose between academic progress and meeting basic needs.

This framing is useful for advocacy because it links financial support to key institutional outcomes: retention, timely completion, and research.

7.6 Interpreting subgroup differences responsibly

Policy makers should interpret subgroup differences responsibly, recognising other influencing features such as household size or childcare. Nevertheless, these comparisons show inequities and justify focused policy support.

Two types within subgroup patterns matter:

- Who pays more for essentials, and
- Who is more vulnerable to insecurity, coping, or crossing burden thresholds?

Both cost-level and vulnerability metrics inform policy, but vulnerability indicators are especially useful for guiding program design, as they highlight where risk is greatest and interventions are most needed.

7.7 What the findings show for advocacy and policy design

The report's findings translate directly into pragmatic implications and policy options, aligned with the GSA's mandate:

- If essential costs are high and rising, fixed stipends become out of sync with student needs. The evidence supports funding changes that are predictable and based on real student cost pressures, rather than on overall inflation.
- Targeted supports for high-burden groups: Indicators such as housing share $\geq 35\%$ and higher food insecurity severity can identify groups for whom targeted grants, emergency funds, and childcare supports are likely to have the highest marginal impact.
- The findings on food and housing insecurity support investing in basic-needs supports, such as expanded food security programs, emergency bursaries, rent deposit assistance, and partnerships with housing providers.
- Survey use can guide service improvements in hours, confidentiality, location, and clarity of eligibility. Service design changes can matter as much as added funding.

The exploratory regression provides a compact, policy-interpretable summary of how cost burden and student circumstances relate to respondents' reported food insecurity. The strongest and most consistent association is the composition of essential spending, particularly housing share, which is linked to substantially higher odds of food insecurity even after accounting for annual funding and other controls. International student status is also associated with markedly higher odds of food insecurity, while higher annual funding is associated with lower odds, confirming the importance of funding adequacy as a basic needs issue. At the same time, these results should be interpreted as associations rather than causal effects because the survey is voluntary and unweighted, and spending measures can capture both monetary strain and differences in resources or living arrangements. Notwithstanding these limitations, the regression is consistent with the descriptive findings and strengthens the report's core message that housing-cost dominance and structural funding gaps are central drivers of basic-needs insecurity for many graduate students.

7.8 How this survey can serve as an ongoing monitoring tool

A major strength of the project resides within its possible repeatability. Researchers can re-estimate the Essential Cost Index, the food insecurity severity score, and threshold-based burden indicators in future cycles to monitor changes over time and evaluate whether policy actions improve student health.

To support this, the discussion should emphasise:

- Keeping core questions stable across waves,
- Adding a small number of carefully chosen new items only when necessary, and
- Report a consistent set of headline indicators each year or term for comparability.

7.9 Summary of the discussion

Overall, this data provides strong, policy-relevant evidence that graduate students experience significant, and sometimes severe, cost pressures related to essentials like housing and food. The findings on cost distribution, insecurity, and coping mechanisms reinforce the requirement for policy steps focused on funding adequacy and targeted basic-needs support.

8. Recommendations and Outputs

8.1 Purpose of the recommendations

The recommendations in this section translate the survey evidence into practical actions that the Graduate Students' Association (GSA) and the University of Alberta can implement or advocate for. The goal is twofold:

- Reduce imminent financial hardship and basic-needs insecurity among graduate students, and
- Establish durable mechanisms for monitoring and responding to graduate-student cost pressures regarding time.

Recommendations are organised by theme and intentionally framed as actionable "outputs" to support policy discussions, budget negotiations, and program design.

8.2 Funding adequacy and cost-of-living alignment

Recommendation 1: Establish a graduate student cost-of-living benchmark for funding decisions. Use the report's Essential Cost Index and category-level expense summaries as a reference basis in discussions about minimum funding expectations. A practical approach is to define a benchmark (e.g., the median essential monthly costs) and track year-over-year changes.
Outputs:

- A one-page benchmark dashboard showing annual changes in the Essential Cost Index and the top expense categories.
- A standard "funding adequacy" slide that links costs to typical stipend structures for advocacy meetings.

Recommendation 2: Advocate for predictable indexing of graduate funding and key student fees. Where funding is fixed or increases slowly, cost pressure compounds. An indexing approach tied to a graduate-student-specific cost indicator, rather than the general CPI, better reflects students' realities.
Outputs:

- A repeatable indicator set suitable for annual reporting (ECI, housing share, food insecurity severity, and high-burden thresholds).
- A short briefing note summarising how these indicators change over time.

Recommendation 3: Expand targeted relief for high-burden students rather than depending exclusively on universal messaging.
The threshold indicators (e.g., housing share ≥ 35%, ECI ≥ P75, higher food insecurity severity) can define qualification or priority targeting for relief funds.
Outputs:

- A "high-burden profile" table that identifies which student groups are most likely to exceed burden thresholds, used to justify targeted programs.

8.3 Food security supports

Recommendation 4: Strengthen food security initiatives using ranked priorities and severity indicators.
Use the ranked initiative questions to identify which supports students prioritise most, and use the severity score to show that food insecurity is not simply "mild" for a subset of respondents.
Outputs:

- A ranked initiatives summary (Any-rank + Borda) to support funding requests.
- A severity distribution figure and subgroup table for inclusion in advocacy materials.

Recommendation 5: Improve the design of access to, not only the supply of, food supports.
If need is high but utilisation (e.g., campus food bank use) is not proportional, focus on barriers such as awareness, hours, stigma, and clarity around eligibility.
Outputs:

- A service-access improvement checklist based on survey findings (e.g., hours, communication, confidentiality).
- A short implementation memo to the relevant campus unit(s) outlining suggested access improvements.

8.4 Housing affordability and housing security

Recommendation 6: Use housing cost and housing-share indicators to support housing-related advocacy.
Housing is typically the dominant essential expense. Housing-share measures translate costs into a simple "budget pressure" story that connects in policy settings.
Outputs:

- A housing dashboard page: housing cost distribution, housing share distribution, and per cent above key thresholds.
- Subgroup comparison visuals (e.g., dependents vs no dependents; residency status) to support equity-based arguments.

Recommendation 7: Align housing supports with reported housing security risks.
Where respondents report specific housing risks (e.g., instability, unaffordable increases, unsafe conditions), supports should handle those risks directly. Likely interventions include emergency rent assistance, rent deposit support, or housing navigation resources.
Outputs:

- A ranked "housing issues" summary and a frequency table of housing security issues.
- A short "housing risks and mitigation options" brief for administrators.

8.5 Transportation and commuting supports

Recommendation 8: Prioritise transportation interventions where commute constraints intersect with cost burden.

Commute time and transportation cost results can support advocacy for transit subsidies, better student access to low-cost parking options, and coordination with transit providers.
Outputs:

- A transport summary page: modes (normalised), commute time distribution, and transport cost by mode.
- A cross-tab figure showing commute time by main mode of transportation.

8.6 Workload and employment pressures

Recommendation 9: Frame financial support as an academic-success issue, not only a welfare issue.
Workload findings help demonstrate that students facing high cost pressure may also be balancing substantial academic demands and paid work. This supports messaging that insufficient funding can affect time-to-degree, research productivity, and retention.
Outputs:

- Summary tables of weekly research and coursework hours and (where available) paid work hours.
- A short narrative slide connecting funding adequacy indicators to academic outcomes (time constraints and stress), grounded in the data.

8.7 Future Survey Cycles

Recommendation 10: Preserve core questions to enable trend tracking.
To make results comparable over time, keep the core modules stable (housing, food, transport, workload, funding adequacy, and key demographics). Limit changes to question wording and response options if required.

Recommendation 11: Add only targeted missing variables if feasible.
If the GSA and University want stronger affordability ratios or more precise subgroup comparisons, future iterations may include items such as approximate monthly income/stipend (after-tax), household size, and whether the reported housing cost is the personal share or the total household cost. These additions should remain minimal to preserve response completion.

Recommendation 12: Formalise an annual reporting timeline.
Run the survey on a consistent schedule (e.g., early fall) and publish a consistent set of headline indicators each cycle. This strengthens advocacy by making the evidence routine and predictable.

8.8 Summary
The survey provides credible, student-centred evidence of essential cost burdens and basic-needs insecurity among respondents. The recommendations emphasise using these indicators to guide funding adequacy discussions, strengthen and target basic-needs supports, and establish a repeatable reporting framework that allows the GSA and University to monitor graduate-student cost pressures regarding time and evaluate whether interventions are working.

9. Conclusion

This capstone project developed and applied a graduate-student–centred approach to measuring cost-of-living pressures at the University of Alberta. Rather than relying on general inflation measures or household cost indices, the project used primary survey data from graduate students to document the necessary expenses and financial limitations that shape graduate student life. The resulting dataset and indicators supply a credible evidence base for both advocacy and institutional planning.

The results show that essential monthly costs, particularly housing and food, represent substantial and recurring obligations for many respondents and are not evenly distributed across the graduate student population. Distributional analyses and derived indicators, including the Essential Cost Index and food insecurity severity measures, show that a meaningful subset of respondents faces high cost burdens and limited flexibility to absorb price increases or unanticipated expenses. Findings related to transportation, commute time, workload, and sources of financial support further demonstrate that monetary strain frequently coincides with time constraints and coping methods that can affect wellbeing and academic progress.

Beyond the descriptive findings, the project's main contribution is a set of repeatable indicators and reporting outputs that the GSA and the University can use to monitor change over time, identify high-burden groups, and support focused interventions. By recording both typical costs and upper-tail experiences where hardship concentrates, the analysis equips decision-makers with clear, interpretable measures to inform funding discussions, student support program design, and broader policy negotiations.

Future survey cycles can be built on this foundation by sustaining core questions to enable trend tracking and by adding a small number of carefully selected items only where necessary to strengthen affordability comparisons and interpretation. With consistent implementation, this survey can function as an ongoing monitoring tool that links graduate students' lived experience to institutional decision-making.

In sum, the project provides the GSA and the University of Alberta with a practical, evidence-based framework to understand graduate student cost pressures, communicate them credibly to stakeholders, and prioritise actions that reduce hardship and support graduate student success.

10. Limitations and reflections

10.1 Limitations of the evidence base

This project presents a clear descriptive picture of graduate-student cost pressures among respondents, with important limitations affecting interpretation and use.

10.1.1 Voluntary response and representativeness.
The survey is a voluntary, non-probability sample, so prevalence estimates (e.g., food insecurity rates) are not precise campus-wide figures. High stress or hardship may have motivated some to respond, while others, encountering constraints, may not have participated. These opposing influences make the direction and magnitude of selection bias unknown from the dataset alone.

10.1.2 Single wave snapshot.
The survey captures a brief window and reflects conditions at a single point in time. Expenses and hardship can fluctuate across the academic year (e.g., moving periods, winter transportation costs) and across economic conditions. The analysis, therefore, provides a baseline rather than a trend, and it cannot estimate within-person changes or causal effects.

10.1.3 Self-reported measures and interpretive variation.
Cost estimates are self-reported and may be subject to recall error, rounding, or different interpretations. For instance, "monthly housing expense" might mean just rent, rent plus utilities, or an individual's share of rent and utilities. Likewise, "transportation costs" may include varied expenditures. Those variations introduce measurement noise and broaden the observed distributions.

10.1.4 Mixed response formats.
The dataset contains categorical responses, numeric entries, multi-select fields, ranking batteries, and free-text responses. In practice, this produces inconsistency in phrasing (e.g., minor variations in labels) and occasional entries that require cleaning rules. Normalisation improves readability and prevents artificial duplication, but it introduces limited analyst judgment in how categories are merged and how rare responses are grouped.

10.1.5 Numeric parsing assumptions.
Where amounts are entered as ranges (e.g., "200–399"), the analysis typically uses midpoints for visualisation and summary statistics. This allows the inclusion of responses that lack overstated precision, but it can smooth variability and obscure within-range differences. Numeric outliers may also reflect data entry mistakes (e.g., extra zeros) or genuinely high costs; without external validation, outlier handling is a balance between plausibility and prudence.

10.1.6 Index construction constraints.
The Essential Cost Index and housing indicators improve interpretability, yet only cover costs measured in the survey. The index is additive and assumes category comparability. It is a proxy for cost pressure but not a full affordability ratio due because of inconsistent income data.

10.1.7 Small subgroup sizes.
Some subgroup analyses, notably by faculty or uncommon programs, use small counts. Small cell sizes can cause unstable estimates and increase re-identification risk. The analysis tackles this by focusing on larger groups and avoiding excessive granularity, though the limitation remains.

10.2. Reflections Regarding Methodological Choices

Several design and analysis decisions were deliberately made to maximise the usefulness of advocacy while continuing analytical integrity.

10.2.1 Prioritising interpretability over complexity.
Given the project's audience and purpose, the analysis emphasised medians, distributions, and threshold indicators that can be explained clearly in policy settings. More complex causal models were intentionally not a focus because the sampling design and cross-sectional structure limit causal interpretation, and because the report's primary use is descriptive evidence for decision-making.

10.2.2 Using severity measures for basic-needs insecurity.
Creating a food insecurity severity score from an agree/disagree question improved interpretability, distinguishing between those with occasional worry and those with multiple severe indicators, thereby increasing policy relevance. This assumes equal weighting and a consistent "agree" interpretation among respondents.

10.2.3 Normalising categories to avoid misleading duplication.
Many survey exports contain minor label differences that can create duplicated categories (e.g., "Full time" vs "Full-time"). Normalising labels is essential for clean visuals and accurate frequency tables, but it requires transparent rules. These rules should be documented and retained for future survey waves to guarantee consistency.

10.2.4 Balancing narrative evidence and privacy.
Open-ended responses provide important context and compelling qualitative evidence, but they also carry a higher risk of identification. The approach taken—summarising themes and using only non-identifying, lightly redacted quotes when necessary—is appropriate for an advocacy-oriented report, but it requires ongoing caution.

10.3 What Would Strengthen Future Iterations

The current survey sets a credible baseline; future cycles could improve exactness and interpretation with small, targeted additions that do not increase respondent burden.

- Income/stipend clarity: A steady monthly take-home income or stipend would enable accurate affordability ratios (expenses as a share of income).
- Household context: household size and whether reported housing costs represent personal share versus total household cost would improve cross-student comparability.
- Location/commute context: a high-level location variable (e.g., living in Edmonton vs outside Edmonton; distance bands) could improve the interpretation of transport and housing trade-offs.
- Repeated measurement: running the survey annually with stable core questions would enable trend evaluation and evaluation of whether policy changes improve outcomes.

10.4 Personal and practical reflections

This project shows the value of combining student-centred measurement with analytical discipline. Cost-of-living experiences are not well captured by general indices: the graduate-student budget is formed by unique constraints. The survey emphasises these constraints in indicators that can be communicated to decision-makers.

Building repeatable infrastructure is important. The long-term value lies in rerunning the analysis pipeline across future survey waves to obtain comparable results, consequently strengthening advocacy through continuous evidence.

The project showcases that the cost of living is linked to wellbeing, inclusion, and academic success, not just budgeting. Framing findings that way, while being transparent about methodological limits, helps keep results both trustworthy and actionable.

References

Alberta Living Wage Network. (2024, November 19). Alberta Living Wage Network releases 2024 living wage rates in partnership with 21 communities. https://www.livingwagealberta.ca/news/alberta-living-wage-network-releases-2024-living-wage-rates-in-partnership-with-21-communities

Alberta Student Aid. (n.d.). Loan limits. Government of Alberta. https://studentaid.alberta.ca/types-of-funding/loan-limits/

Canadian Alliance of Student Associations. (2023, August 16). Living in the red: An examination of food and housing challenges facing Canadian post-secondary students. https://www.casa-acae.com/living_in_the_red

City of Edmonton. (2026). ETS fares & passes. https://www.edmonton.ca/ets/fares-passes

Elgar, F. J., Pickett, W., Pförtner, T.-K., Gariépy, G., Gordon, D., Georgiades, K., Davison, C., Hammami, N., MacNeil, A. H., & others. (2023). Relative food insecurity, mental health, and wellbeing in Canada. SSM – Population Health, 22, 101343. https://doi.org/10.1016/j.ssmph.2023.101343

Laframboise, S. J., Bailey, T., Dang, A.-T., Rose, M., Zhou, Z., Berg, M. D., Holland, S., Abdul, S. A., O'Connor, K., El-Sahli, S., Boucher, D. M., Fairman, G., Deng, J., Shaw, K., Noblett, N., D'Addario, A., Empey, M., & Sinclair, K. (2023). Analysis of financial challenges faced by graduate students in Canada. Biochemistry and Cell Biology, 101(4), 326–360. https://doi.org/10.1139/bcb-2023-0021

Ronson, D. (2025, September 9). Student housing crunch eases. University Affairs. https://universityaffairs.ca/news/student-housing-crunch-eases/

Statistics Canada. (2022, September 7). Tuition fees for degree programs, 2022/2023 [Archived report]. Government of Canada. https://www150.statcan.gc.ca/n1/daily-quotidien/220907/dq220907b-eng.htm

Statistics Canada. (2024, November 19). Housing challenges related to affordability, adequacy, condition and discrimination, August 2 to September 15, 2024. https://www150.statcan.gc.ca/n1/daily-quotidien/241119/dq241119b-eng.htm

Statistics Canada. (2024, September 10). Tuition in Canada: Modest increases and widening gaps, 2024/2025. https://www150.statcan.gc.ca/n1/daily-quotidien/240910/dq240910c-eng.htm

Statistics Canada. (2025, May 21). Study: Tuition fees for degree programs, 2024/2025. Government of Canada. https://www150.statcan.gc.ca/n1/daily-quotidien/250521/dq250521a-eng.htm

Statistics Canada. (2025, September 10). Daily — Postsecondary enrolments, 2023/2024. Government of Canada. https://www150.statcan.gc.ca/n1/daily-quotidien/250910/dq250910d-eng.htm

Statistics Canada. (2025, September 10). Tuition in Canada: Modest increases and widening gaps, 2025/2026. https://www150.statcan.gc.ca/n1/daily-quotidien/250910/dq250910d-eng.htm

University of Alberta. (2024). Minimum guaranteed funding for PhD students. https://www.ualberta.ca/en/graduate-studies/fees-funding/other-funding-support/minimum-guaranteed-funding-for-phd-students.html

University of Alberta. (2024). Tuition and fees. https://www.ualberta.ca/en/admissions/tuition-and-fees/index.html

Uppal, S. (2023). Food insecurity among Canadian families (Insights on Canadian Society, Statistics Canada Catalogue No. 75-006-X 2023001). Statistics Canada. https://www150.statcan.gc.ca/n1/pub/75-006-x/2023001/article/00013-eng.htm

Wang, Y., Fafard St-Germain, A.-A., & Tarasuk, V. (2023). Prevalence and sociodemographic correlates of food insecurity among post-secondary students and non-students of similar age in Canada. BMC Public Health, 23(1), 954. https://doi.org/10.1186/s12889-023-15756-y